Zhang 2019 Follicular Lymphoma — analysis summary

RUN 4 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony θ-sweep (5→20) + per-neighbourhood iLISI: mixing plateaus at θ=8 (same-sample nbr frac 0.97→0.46; iLISI sample 1.9/4, patient 1.5/2). Single connected kNN component; 0 detached islands. Patient-private FL clones integrate into one continuous manifold; signature geography is cell-cycle-independent.

Gate 2 — module geography → GO (non-branching)

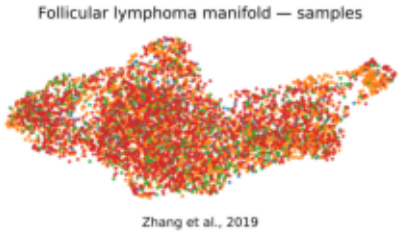
Non-branching case: Gast/pSom overlap on one axis (spatial r=−0.19, no two competing arms); Crest an inverse/terminal gradient (Crest vs Gast −0.52, vs pSom −0.60; joint pSom&Crest-high enrichment 0.23 < 1). Single Gast/pSom → Crest developmental axis present → GO.

Gate 3 — hub cross-repression → NA

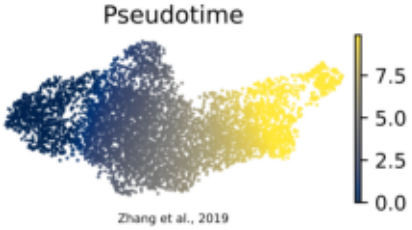
No bifurcation (curve, 0 forks) → branching-only gate not applicable. 04_correlation / 05_modules not produced.

STEP FIGURES

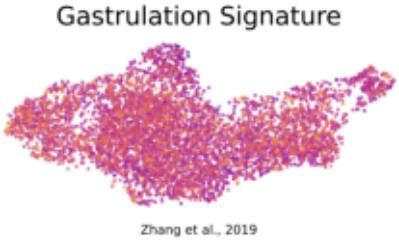
Manifold — samples



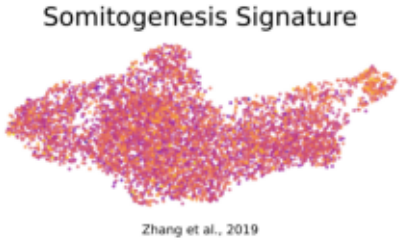
Pseudotime



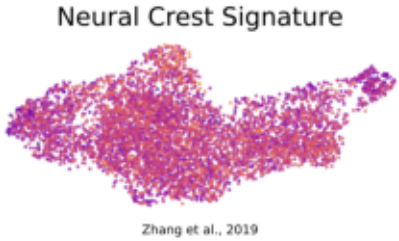
Gastrulation



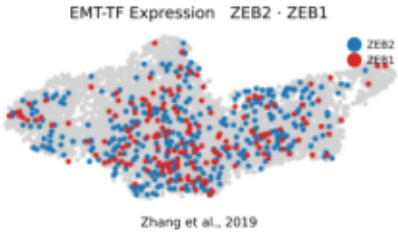
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Follicular lymphoma malignant B cells (Zhang et al., 2019), 10x — 5,645 malignant cells · 4 samples · 2 patients (FL1018, FL2001; FL1018T2 transformed-DLBCL); one continuous gastrulation → neural-crest axis, no arm split.

Verdict.
Pattern REPRODUCES as the non-branching case — Gate 1 GO / Gate 2 GO / Gate 3 NA (linear). No cross-repression hub. Module means (raw) Gast +0.012 / pSom +0.006 / Crest +0.009. EMT-TFs ZEB2 (27%) · ZEB1 (12%), both terminus-associated; no root-end TF.

Parameter choices

Sample threshold	≥100 cells → all 4 kept (5,645 cells); min sample 680 ≈ 2×floor, no under-populated sample	**Overdispersed genes**	636 (scFates find_overdispersed, GAM residuals)
n_pcs	30 (variance-ratio elbow past PC20; floored to robust 10x range)	**Integration**	Harmony θ=8 on sample (mixing plateau across θ=5–20; FL clones patient-private)
Embedding	k=30, UMAP spread=1.3; radius-graph island removal (0 removed)	**Geometry**	Linear (non-branching); scFates curve N=20, 0 forks, fold-free (SA 2.96); PPT spurious forks
Root	Gast/pSom-high tip (Crest-low); order Gast t≈0.04 → Crest 0.59 → pSom 0.82	**EMT-TFs**	ZEB2 (27%) · ZEB1 (12%) both terminus-associated; no root-end TF; VIM excluded